1. I'm using β for thermal factor so we'll use $H(p) = \alpha p^b$

$$\begin{aligned} Z &= \frac{1}{N! h^{DN}} \int d^{DN}q d^{DN}p e^{-\beta H(p)} \\ &= \frac{V^N}{N! h^{DN}} \left(\int d^D p e^{-\alpha \beta p^b} \right)^N \end{aligned}$$

We want to get the integral out as a constant that we can take the derivative of. Using Helmholtz free energy and $Z = Z(V) Z_C$

$$\begin{aligned} F &= -\frac{\ln Z}{\beta} = -\frac{\ln(Z(V)) + \ln(Z_C)}{\beta} \\ p &= -\left(\frac{\partial F}{\partial V} \right)_{T,N} = \frac{1}{\beta} \left(\frac{\partial Z(V)}{\partial V} \right)_{T,N} \end{aligned}$$

We don't actually have to make the integral dimensionless, it already doesn't have volume dependence.

$$\begin{aligned} p = \frac{1}{\beta} \frac{\partial}{\partial V} \left[\ln V^N + \ln(C) \right] = kTN \frac{\partial}{\partial V} \ln V \\ pV = NkT \end{aligned}$$

2.

First lets get $Z_1 = Z_K Z_U$. In the high-T limit β gets very small so we can just use the first terms of $e^{-U(r)\beta} = 1 - \beta U(r) + \frac{1}{2}\beta^2 U^2 - \ldots$. We'll hang on to β^2 since there are some logs and derivatives coming up. Then replace $U(r) \to U = \langle U(r) \rangle$

$$\begin{aligned} Z_U &= \int d^3 r \left(1 - \beta U + \frac{1}{2}\beta^2 U^2 \right) \\ &= V \left(1 - \beta U + \frac{1}{2}\beta^2 U^2 \right) \\ Z_K &= \frac{1}{h^3} \int d^{3N} p e^{K\beta} = \frac{1}{\lambda^3} \\ Z_1 &= Z_{\text{ideal}} \left(1 - \beta U + \frac{1}{2}\beta^2 U^2 \right) \\ Z_1 &= Z_{\text{ideal}} Z_{\text{corr}} \end{aligned}$$

Get to heat capacity from $\ln(Z) \approx N \ln(Z_1/N) + N$.

$$
\begin{aligned}
C_V &= k\beta^2 \frac{\partial^2 \ln Z}{\partial \beta^2} \\
&= k\beta^2 \frac{\partial^2}{\partial \beta^2} \left[N \ln\left(\frac{Z_1}{N}\right) + N \right] \\
&= k\beta^2 N \frac{\partial^2}{\partial \beta^2} \left[\ln\left(\frac{Z_{\text{ideal}}}{N}\right) + \ln\left(Z_{\text{corr}}\right) \right] \\
&= C_{\text{ideal}} + k\beta^2 N \frac{\partial^2}{\partial \beta^2} \ln\left(Z_{\text{corr}}\right)
\end{aligned}
$$

One chain rule later...

In [28]:
```
from sympy import *
N, beta = symbols('N beta', real=True, positive=True)
U = symbols('U', real=True)
Zcorr = 1 - beta*U + Rational(1,2) * beta**2 * U**2
diff(ln(Zcorr), beta, 2).simplify()
```

Out[28]:
$$\frac{2U^2\left(U^2\beta^2 - 2U\beta - 2\left(U\beta - 1\right)^2 + 2\right)}{\left(U^2\beta^2 - 2U\beta + 2\right)^2}$$

wait wait nevermind. Use $\ln(1+x) \approx x - x^2/2$ since $\beta \to 0$

In [29]:
```
x = Zcorr-1
display(x - x**2/2)
```

$$\frac{U^2\beta^2}{2} - U\beta - \frac{\left(\frac{U^2\beta^2}{2} - U\beta\right)^2}{2}$$

Sending higher order terms to zero, we'll just use $\ln(Z_{\text{corr}}) \approx -U\beta + U^2\beta^2/2$

$$C_V = C_{\text{ideal}} + Nk\beta^2U^2$$

which still vanishes quickly.

3.

the volume integral is

$$A = V/H$$

$$Z_U = A\int_0^H e^{-mgz\beta} = \frac{V}{\beta mgH}\left(1-e^{-mgH\beta}\right)$$

$$Z_N = \frac{1}{N!\lambda^{3N}}Z_U^N = \frac{V^N}{N!\lambda^{3N}}\left(\frac{1-e^{-mgH\beta}}{mgH\beta}\right)^N = Z_{N,\text{ideal}}Z_{\text{corr}}$$

$$C_{\text{corr}} = k\beta^2\frac{\partial^2}{\partial\beta^2}\ln Z_{\text{corr}}$$

$$= k\beta^2 N\frac{\partial^2}{\partial\beta^2}\left[\ln\left(1-e^{-mgH\beta}\right)-\ln\beta\right]$$

In [43]:
```
m,g,H = symbols('m g H', real=True, positive=True)
G = m*g*H

Zcorr1 = (1-exp(-G*beta))/(G*beta)
diff(ln(1-exp(-G*beta)), beta, 2).simplify() - diff(ln(beta), beta, 2).simplify()
```

Out[43]:
$$-\frac{H^2g^2m^2}{4\sinh^2\left(\frac{H\beta gm}{2}\right)}+\frac{1}{\beta^2}$$

$$C_{\text{corr}} = kN\left[1-\left(\frac{mgH\beta}{2\sinh\left(\frac{mgH\beta}{2}\right)}\right)^2\right]$$

$$F_{\text{corr}} = -\ln Z_{\text{corr}}/\beta = -\frac{N}{\beta}\ln\left(\frac{1-e^{-mgH\beta}}{mgH\beta}\right)$$

Note that $\sinh(x\to\infty)\to\infty$ very fast and $\lim_{x\to 0}x^2/\sinh^2(x)\to 1-x^2/3$.

$$C_{\text{corr}}(\beta\to 0) = \frac{kN}{12}(mgH\beta)^2$$
$$C_{\text{corr}}(\beta\to\infty) = 0$$
$$F_{\text{corr}}(\beta\to 0) = \dots$$
$$F_{\text{corr}}(\beta\to\infty) = 0$$

Using $\frac{1-e^x}{x} = 1-x/2+x^2/6$, we can use the friendly expansion of $\ln$ near zero

$$\ln\left(\frac{1-e^x}{x}\right) = -\frac{x}{2}+\frac{x^2}{24}$$

$$F_{\text{corr}}(\beta\to 0) = \frac{NmgH}{2}-\cancel{\beta\frac{N(mgH)^2}{24}}$$

Which makes sense since that is the average energy of uniformly distributed particles

At $\beta<\beta_c$ the gas escapes. Spherical coordinates.

$$Z_1 = Z_K Z_U$$
$$Z_U = 4\pi \int_R^\infty e^{-\beta U_0 \ln(r/R)} r^2 dr$$
$$= 4\pi R^3 \int_1^\infty x^{2-\beta U_0} dx$$

The integral converges for anything faster than x^{-1} so we have

$$2 - \beta_c U_0 = -1$$
$$\beta_c = \frac{3}{U_0} \to T_c = \frac{U_0}{3k}$$

Interesting how integral divergence is computed by gasses in statistical mechanics. I don't think volume exists here. Above β_c we have

In [46]:
```
x, a, U0 = symbols('x a U_0', positive=True, real=True)
sol = integrate(x**(-1-a), (x,1, oo))
sol.subs(a, beta*U0-3) # 2-beta U0 = -1 - a -> a = beta U0 - 3
```

Out[46]:
$$\frac{1}{U_0\beta - 3}$$

$$Z_U = \frac{4\pi R^3}{U_0\beta - 3}$$
$$Z_N = \frac{1}{N!\lambda^{3N}} \left(\frac{4\pi R^3}{U_0\beta - 3} \right)^N$$
$$F = -\frac{N}{\beta} \ln(Z_U/\lambda^3 N) + N$$
$$C_V = k\beta^2 N \frac{\partial^2}{\partial \beta^2} \ln\left(\frac{1}{U_0\beta - 3} \right) = N \frac{k\beta^2 U_0^2}{(U_0\beta - 3)^2}$$

which goes to infinity for $\beta \to 3$

5. No?

σ_{ik} is the gradient in electric field and σ_{ll} is zero aside from the point source. $U\beta \ll 1$.